

Session Objectives

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Medicine

- Define primary and secondary types of headaches
- Distinguish between primary and secondary headaches according to the international classification of headache disorders
- Identify different types of headaches based on their presentation, symptoms, and causes
- Identify “Red Flags” in the evaluation of headaches
- List treatment options based on the type of headache

<https://exchange.scholarrx.com/brick/m645g1q65jq1>

<https://exchange.scholarrx.com/brick/secondary-headaches>

Headache: A Disabling Problem

The Burden of Headache

Symptom Burdens

During Attacks

- Pain & suffering leading to reduced functional ability

Between Attacks

- Fear of next attack leading to avoidance behaviors and lifestyle compromise

Secondary Burdens

During Attacks

- Loss of work-time/work-effectiveness
- Failed or abdicated social roles
- Curtailed social and leisure activities

Beyond Attacks

- Failure to meet full potential
- Cessation of social planning
- Breakdown of relationships

Primary Headaches

Migraine

- With Aura
- Without Aura

Tension-Type Headache

- Episodic
- Chronic

Cluster Headache and Other Trigeminal Autonomic Cephalalgias

- Cluster headache
- Paroxysmal hemicrania
- Short unilateral neuralgiform headache with conjunctival injection and tearing (SUNCT)
- Short-lasting unilateral neuralgiform headache attacks with autonomic symptoms (SUNA)
- Hemicrania continua

Secondary Headaches:

- IHS Classification

Headache attributed to:

- Head and/or neck trauma
- Cranial or cervical vascular disorders
- Nonvascular intracranial disorders
- Substance or withdrawal thereof
- Infection
- Disorder of homeostasis
- Disorder of cranium, neck, eyes, ears, nose, sinuses, teeth, other facial structures
- Psychiatric

Headache Assessment

Purpose of assessment

- Exclude any urgent, emergent, and/or serious cause
- Make an accurate diagnosis of headache type
- Identify exacerbating/aggravating factors
- Optimize treatment (acute and preventative)
- Explain diagnosis and rationale for treatment

Assessment Tools

- History taking
- Physical exam
- Headache diary
- Imaging studies
- Laboratory analysis

Headache Assessment

History Taking - Key Questions

Time

- Onset, frequency, patterns, duration

Character

- Description, intensity, nature of pain

Location

- Localized, diffuse, radiating, laterality

Causes

- Predisposing, triggering, exacerbating, receiving factors

Response

- Patient's actions and limitations during headache
- Medications used and how often

Associated symptoms

- Auras, nausea, vomiting, visual changes, photo/phonophobia, lacrimation, jaw pain

Headache History

- Does the patient take any medications?
 - Prescription, alternative/herbal, over-the-counter
- Does the patient have any other medical conditions?
 - HIV, cancer, hypertension, diabetes, hypothyroidism, seizures, etc.
- Surgical history
- Allergies
- Social history
 - Smoking, EtOH use, drug use, caffeine use
 - Stress/anxiety level
 - Sleeping habits
 - Eating habits
 - Exercise habits

Headache Triggers

- Diet
 - EtOH
 - Chocolate
 - Aged cheese
 - MSG
 - Aspartame
 - Caffeine
 - Nuts
 - Low H2O intake
- Hormones
 - Menses
 - Ovulation
 - Oral contraceptives
 - Hormone replacement therapy
- Stress/Anxiety
 - Loss or change:
 - Death
 - Divorce
 - Moving/relocation
 - Crisis period
 - Intense activity
 - Let-down period
- Change in habits
 - Weather
 - Travel
 - Diet/sleep
 - Physical activity

Migraine Headache: Aura

- Appears over 5 to 20 minutes and usually lasts less than 60 minutes
- Aura symptoms can be visual, sensory, or motor in nature
- Visual aura is the most common type of neurological event (90%)

Migraine Headache: Headache/Pain

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- Typically unilateral, throbbing, moderate to severe pain, and aggravated by physical activity
- Usually gradual onset
- Pain usually peaks and then subsides:
 - 4 to 72 hours in adults
 - 1 to 48 hours in children
- Usually accompanied by nausea (90%) and vomiting (30%)
- Photophobia, Phonophobia, Osmophobia

Migraine Headache: Postdrome

- Tired, listless, irritable, mood changes, scalp tenderness
- Some patients will feel unusually refreshed or euphoric, whereas others report depression or malaise
- Often, some symptoms of the headache phase can continue

Migraine Headache: Without Aura

Diagnostic Criteria

Attacks lasting 4-72 hours, having occurred at least 5 times

At least *two* of the following:

- Unilateral
- Pulsating
- Moderate to severe
- Aggravated by exercise

At least *one* of the following:

- Nausea
- Photophobia
- Phonophobia

Migraine Headache: With Aura

Diagnostic Criteria

- Migraine without aura PLUS
 - One or more transient focal neurological signs:
 - Visual
 - Sensory
 - Aphasia
 - Motor
- Gradual development of aura symptoms
- Aura lasts 4 to 60 minutes followed by or accompanied by headache within 60 minutes

Migraine Headache

- Predisposing Factors/Triggers -

- Family history
- H/o travel sickness
- H/o cyclical vomiting in childhood
- Stress
- Menstruation
- Hormonal replacement therapy
- Oral contraceptive pills
- Head/neck trauma
- Strenuous exercise
- Relaxation after stressful events
- Change of habits
 - Missed meals, missed sleep
- Bright lights
- Loud noises
- Smells
 - Perfumes, gasoline
- Dietary (<10%)
 - Chocolate
 - Cheese
 - Aspartame
 - EtOH

Migraine Headache

- Acute Treatment -

Basic Principles

- Correct dose
- Correct formulation
- Step-up treatment
 - Simple analgesia —> combinations —> triptans/ergots
- Early intervention
- Use at least 3 times before judging effectiveness
- Use a maximum of 2-3 days per week or 8 days in a month

Migraine Headache

- Acute Treatment -

- Analgesics
 - Acetaminophen
 - NSAIDs
- Specific anti-migraine therapies
 - Triptans (5HT- agonist (sumatriptan, Rizatriptan, etc.)
 - Ergots (Ergotamine)
- Anti-emetics
 - Promethazine (Phenergan)
 - Prochlorperazine (Compazine)
 - Metoclopramide
- Combinations
 - Excedrin (Acetaminophen/aspirin/caffeine)



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Migraine Headache

- Management and Prevention -

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- Lifestyle modifications
 - Regular sleep
 - Regular meals
 - Moderate amount of exercise
 - Drink plenty of water
 - Limit caffeine, alcohol, and other drugs
 - Reduce stress
- Cognitive behavioral therapy
- Biofeedback
- Relaxation training
- OMT
- Acupuncture
- Pharmacological treatments

Migraine Headache

- Management and Prevention -

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When to Use

- Frequent headaches - >3-4/month
- Migraine significantly interferes with patients daily life despite abortive treatment
- Acute/abortive therapies contraindicated, ineffective, not tolerated, or OVERUSED

Migraine Headache

- General Principles -

- Consider comorbidities and lifestyle factors
- Titrate dose - “start low and go slow”
- Aim for optimal dose
- Try medication for at least 3 months
- Some patients may relapse and require long-term therapy

Tension-Type Headache

- **Most common type of headache!**
- Women > men
- Mild to moderate in severity
- Genetics may play a minor role
- Lasting as little as 30 minutes to as long as 7 days
- Myofascial tenderness is increased
- Often coexists with migraine headache
- Triggers - stress, fatigue, lack of sleep
- Comorbid conditions - depression and anxiety in >50% of patients

Tension-Type Headache

- IHS Criteria -

- Require at least 10 episodes of headache, each lasting 30 minutes to seven days, which fulfill the following conditions:

At least two of the following:

- Bilateral location
- Pressing or tightening (non-pulsating) quality
- Mild or moderate intensity
- Not aggravated by routine physical activity such as walking or climbing stairs

Both of the following:

- No nausea or vomiting
- No more than one of photophobia or phonophobia

Tension-Type Headache

Episodic Tension-Type Headache

- Episodic TTH (*<15 headaches a month*) - initial treatment with simple analgesics
 - NSAIDs or Aspirin
 - Acetaminophen
 - Parenteral Medications
 - Ketorolac, Metoclopramide, Chlorpromazine
 - Limited Evidence - used in severe cases

Chronic Tension-Type Headache

- Chronic TTH (*≥15 headache days a month*), *for at least 6 months*
 - Acute symptoms - NSAIDs or acetaminophen
 - Amitriptyline, skeletal muscle relaxants, fluoxetine
 - Stress managements, OMT, physical therapy, massage, acupuncture

Cluster Headache

- Age of onset 25-50 years, Male > Female
- Diagnostic criteria:
 - Attacks “clustered” in time (>5)
 - Severe unilateral, orbital supraorbital, and/or temporal pain
 - Duration of 15 to 180 minutes, occurring 1 to 8 times per day
 - Associated with one of the following ipsilateral signs:
 - Conjunctival injection/lacrimation
 - Nasal congestion/rhinorrhea
 - Eyelid edema
 - Forehead/facial swelling
 - Miosis/ptosis (possibly both)
 - Restlessness/agitation

Secondary Headaches

- Meningitis or Encephalitis -

- Combination of new headache (87%), neck stiffness (83%), fever (77%), and altered mental state (69%)
- Infection involving brain parenchyma (encephalitis) presents similarly with headache (81%), fever (90%), and altered consciousness (97%), but seizures are frequent (67% in encephalitis while only 5% in meningitis).
- Pain is caused by inflammation of intracranial pain-sensitive structures including blood vessels at the base of the brain.
- Commonly throbbing in character, bilateral, and occipital or nuchal in location.

Secondary Headaches

- Giant Cell (Temporal) Arteritis -

- Systemic vasculitis that affects medium-sized and large arteries (temporal), especially branches of the external carotid artery
- Inflammation of the pain-sensitive arterial wall produces headache, and arterial stenosis leads to ischemia
- Women > men and is uncommon before 60 years of age
- Most common symptom is a new, nonspecific headache, unilateral or bilateral and is often fairly severe and boring in quality
- ESR is elevated, but not invariably; requires prompt evaluation to avoid vision loss
- Diagnosis made by biopsy of an affected temporal artery, which is characteristically thickened, nonpulsatile, dilated, and tender

In Summary...

- Define primary and secondary types of headaches
- Distinguish between primary and secondary headaches according to the International Classification of Headache Disorders and International Headache Society
- Identify the different types of headaches based on their presentation/symptoms/causes
- Identify “RED FLAGS” in the evaluation of headaches
- List treatment options based on type of headache